PriorityKV: Structure-Aware KV Retention with a Measurable Operating Boundary

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Abstract

Serving a long agent conversation means storing a key–value (KV) cache whose size grows with the transcript, and shrinking that cache means deciding which positions to discard. Existing selectors read the model's own behaviour: accumulated attention, or attention within an observation window. A chat transcript, however, already carries a second signal that costs nothing to compute—its message roles, tool contracts, and constraint markers—and it is natural to ask whether that free signal is sufficient. This paper answers *when*.

We measure the operating range of a structural retention prior as a dose–response curve rather than inferring it from endpoints. Sweeping the protected-role mass of a controlled agent benchmark across eight levels (5.2%–92.2%), crossed with three keep budgets, over 15,360 scored generations on Qwen3-8B and Llama-3.1-8B, we find a sharp boundary: structural retention is at ceiling while the protected set fits the budget and degrades immediately once the oversubscription ratio $|S \cup M|/B$ exceeds approximately 1.1. The boundary replicates across both model families. Past it, the ordering against attention-based selection reverses: structure loses to matched-budget SnapKV in 20 of 24 Llama cells, winning no discordant pair in any of them.

Pooling across task types conceals the mechanism, and the exception identifies it. At 18× oversubscription, tool-schema conformance remains at 1.000 while instruction supersession falls to exactly 0.000. Because the evaluated policy resolves ties by position, state carried in the leading system block survives arbitrary oversubscription, whereas state accumulated across mid-conversation turns does not. The governing quantity is therefore not protected mass alone but its interaction with where load-bearing state sits relative to the tiebreak—a distinction that predicts, rather than retrospectively excuses, the failure of the same prior on externally authored agent traces.

We use that mechanism to make a falsifiable external prediction. A blended policy whose single weight is read from the prompt with nothing fitted never loses to SnapKV across 48 cells and significantly exceeds it on Qwen at a 2% budget ($p = 4.9 \times 10^{-4}$), but the same mechanism predicts no benefit on the Berkeley Function Calling Leaderboard, whose failures are dominated by loss of multi-turn state rather than of schemas. We preregistered that prediction, and the test confirms it: over 233 paired conversations ADAPT is indistinguishable from SnapKV ($\Delta = -0.026$, $p = 0.33$), inside the bound the registration specified. The method therefore confers no external benefit, while the mechanism that predicts its failure does have external predictive power. We additionally report a matched-precision storage study as an explicit negative, and document two experimental iterations that returned nulls, each for an identified cause.

1 Introduction

Autoregressive inference stores the keys and values of every prior position so that each decoding step can attend over the prefix without recomputation. The resulting cache grows linearly with sequence length and, in agent workloads where transcripts accumulate over many turns, becomes the binding constraint on serving capacity [1, 2]. Systems respond by paging the cache, by restricting attention to a sliding window, by evicting positions judged unimportant, or by storing them at reduced precision [3, 4, 5, 6, 7].

Eviction requires a criterion, and the criteria in use are derived from the model. H2O accumulates attention mass per position; SnapKV scores prefix positions by attention received from an observation window near the end of the prompt; PyramidKV varies the per-layer budget according to how information funnels through depth. These signals are informative precisely because they reflect what the model is doing, but obtaining them is not free: they require attention statistics that exist only after the relevant computation has been performed.

A chat transcript carries a different signal, and this is the observation that motivates the present work. Its serialization already distinguishes a system preamble from a user turn, a tool schema from a tool result, an

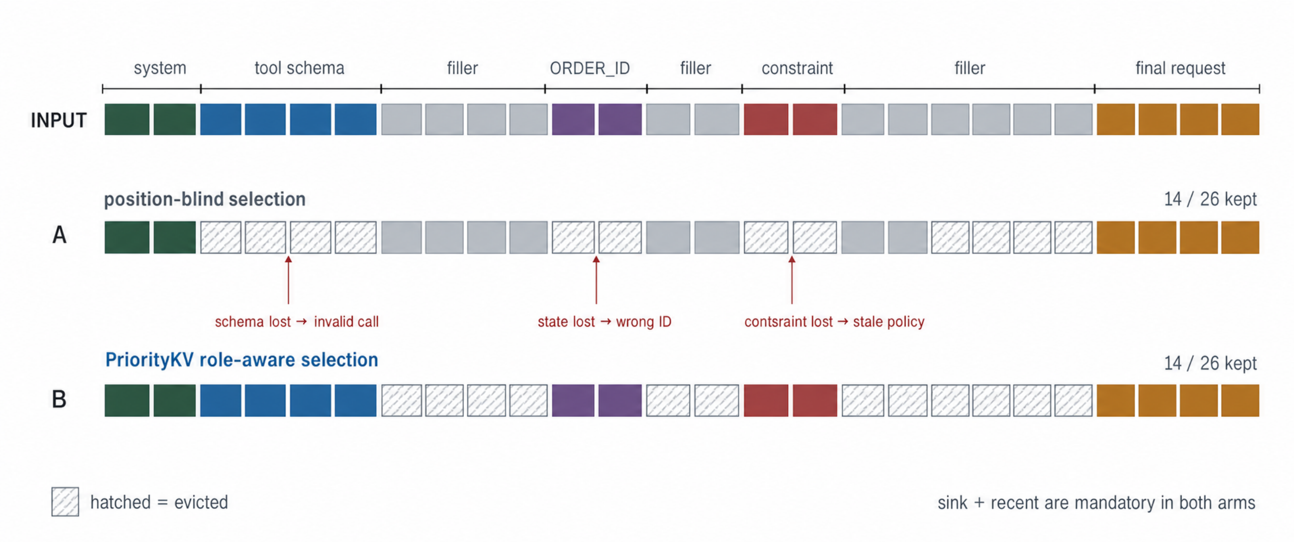

Figure 1: Equal storage cost, unequal failure cost. Two policies keep the same illustrative 14-of-26-token budget. Hatched cells are evicted; callouts show the resulting failure modes. The figure is schematic and reports no empirical counts.

instruction from filler. That structure is available from the prompt alone, before any computation, at negligible cost. If a retention rule built on it were adequate, a serving system could make eviction decisions without paying for attention statistics at all. The question is not whether such a rule is better than attention-based selection—we will show it generally is not—but whether there exists a characterisable regime in which the cheap signal suffices, and whether that regime can be recognised in advance.

Structural retention itself is established. RoleKV manages agent KV blocks by semantic role rather than recency, though losslessly, deciding placement in the memory hierarchy rather than discarding [12]; work on structural-role bias allocates budget across document-structural categories and reports that the achievable gain is bounded by role density, becoming indistinguishable from SnapKV where the informative role is sparse [13]. What is not established is the behaviour at the opposite extreme. Agent traces do not suffer from too few protected tokens; they suffer from too many, because tool schemas can occupy nearly the whole context. Prior work in this specific line, including an earlier version of this study, asserted a boundary there on the basis of two observations: a synthetic benchmark on which structural retention performed well and carried little protected content, and an external benchmark on which it failed and carried a great deal. Two points do not establish a curve, and a boundary inferred from the endpoints of an interval cannot distinguish a real threshold from a coincidence of two unrelated effects.

Thesis. This paper replaces that inference with a measurement, and in doing so arrives at a mechanism different from the one previously proposed. Structural retention has a sharp operating boundary located where the protected set ceases to fit the keep budget; the boundary is reproducible across model families and computable before generation. But the ratio alone does not determine what is lost when the boundary is crossed. Because the evaluated policy breaks ties by position, state carried in the leading system block is retained at any degree of oversubscription while state accumulated across mid-conversation turns is discarded entirely. It is this second factor, not the ratio, that determines whether a workload is served well by the cheap signal.

Contributions.

1. **A measured operating range.** We introduce a sweep methodology that parameterises the protected-role mass of a controlled agent benchmark while holding the task, its scoring, and its context length fixed, and we report the resulting dose–response curve over eight mass levels, three keep budgets, six retention arms and two model families (Sections 5.1–5.2).

2. **The mechanism.** We dissociate task families to show that a position-ordered tiebreak, not protected mass, decides which state survives oversubscription, and we show the dissociation replicates across architectures (Section 5.3).

3. **A falsifiable external prediction, registered and confirmed.** We derive from the mechanism a directional prediction for an externally authored benchmark and pre-register it—with a budget-selection rule and a power analysis—before the experiment is scored; the test then returns the predicted null ($\Delta = -0.026$, $p = 0.33$, $n = 233$ paired), establishing external predictive power for the mechanism while confirming that the method itself does not transfer (Section 6).

4. **Negative and null results reported in full.** We report a storage study that finds no quality separation, two sweep iterations that returned nulls with identified causes, and a failure-mode census showing that the external benchmark is limited more by model capability than by retention (Sections 7, 5.1, 6.2).

2 Background and Related Work

Paging and streaming. PagedAttention separates logical KV blocks from non-contiguous physical storage to reduce fragmentation and enable sharing across requests [2]; it manages placement rather than selecting which positions deserve to remain. StreamingLLM observes that a small number of initial tokens act as attention sinks and that retaining them alongside a recent window supports unbounded streaming [3]. We adopt sink-plus-recent as a mandatory region for every policy and as the shape of our position-only control, without claiming to reproduce the streaming setting in which it was introduced.

Attention-derived eviction. H2O retains heavy hitters by accumulated attention together with recent positions [4]. SnapKV selects prefix positions using attention from an observation window at the end of the prompt, exploiting the empirical stability of those patterns [5]. PyramidKV allocates budget non-uniformly across layers [6]. All three infer importance from model behaviour. We use these methods as the standard against which the sufficiency of a behaviour-free signal is judged. Except where noted, the attention comparison in this paper uses the `kvpress` implementation of SnapKV rather than a reimplementation.

Role- and structure-aware retention. The idea that a token's structural role should inform its retention is not new, and two lines of prior work bear directly on ours. RoleKV tags KV blocks with agent-workflow roles—system, instruction, reasoning, action, tool output—and manages them by role rather than recency, motivated by an inverted age–importance effect in which the oldest blocks remain the most attended [12]. RoleKV is, however, a *lossless* policy: it decides where blocks reside in the memory hierarchy and does not alter model outputs. The present work studies lossy eviction, where a discarded position cannot be recovered and the failure modes are correspondingly different.

Closer to our question, work on structural-role bias applies a role-conditional budget allocator over document-structural categories such as schema labels, table headers and prose, and reports that the achievable gain is *bounded by role density*: where the informative role occupies a small fraction of the context, the policy becomes indistinguishable from SnapKV [13]. That result establishes a boundary at the *sparse* end of the role-density axis. Our contribution is the opposite end. We show that when protected content instead saturates the budget, the policy does not merely lose its advantage but inverts its ordering against attention-based selection, and—the finding we regard as new—that *which* state is lost is determined not by density but by the position-ordered tiebreak the policy falls back on (Section 5.3). The two results are complementary descriptions of when a structural prior is usable.

Diagnostics for when eviction helps. A related line asks not which selector is best but when eviction succeeds at all. A fixed-contract diagnostic decomposes eviction failure into access estimation, value ranking, and score-to-cache projection, and derives a pre-generation margin that predicts whether value-aware selection will help [14]. That study evaluates the same two 8B models used here and explicitly confines itself to attention- and value-aware selection, leaving role-based and structural retention unexamined. The diagnostic we report—the oversubscription ratio together with the location of load-bearing state—is the structural counterpart it does not cover.

Quantization. KIVI shows that asymmetric group-wise low-bit quantization of the KV cache preserves quality while keeping a residual of recent tokens at full precision [7]. Our storage study is not a new quantizer and is not positioned against KIVI; it asks the narrower question of whether *role-aware placement* of a fixed INT4 fraction outperforms uniform placement, and answers in the negative. FlashInfer supplies the composable kernels and log-sum-exp state merging that our mixed-precision decode path uses [8].

Evaluation. LongBench and RULER assess long-context capability broadly [9, 10]; our benchmark instead isolates failure attribution in synthetic agent protocols and cannot substitute for them. BFCL provides stateful multi-turn function-calling tasks with an executable checker [11], and τ-bench evaluates policy-constrained tool interaction with a simulated user [15]. We use BFCL for external task success and public τ-bench trajectories for a generation-free retention audit; neither informed the construction of our benchmark.

3 Retention Policies and the Oversubscription Boundary

3.1 Matched-budget retention

Let a tokenised trace be $x_{1:n}$ with structural roles $r_{1:n}$ assigned by a tagger that reads message roles and transparent schema or constraint markers. The tagger inspects neither benchmark answers nor future attention. For keep fraction k the nominal budget is

$$B(n,k) = \mathrm{round}(kn), \tag{1}$$

and the mandatory set is $M = M_{\mathrm{sink}} \cup M_{\mathrm{recent}}$, holding the first 16 and last 128 positions. Every compressed arm selects a retained set $A \supseteq M$ with

$$|A| = \max\big(B(n,k),\, |M|\big), \tag{2}$$

so that arms differ only in *which* positions survive, never in how many. Position-blind retention fills from the recent edge. Structural retention first admits system, tool, constraint and other state-bearing positions, then fills any residual budget from the recent edge.

3.2 The boundary, and why the tiebreak matters

Write S for the structurally protected positions. If $|S \cup M| \le B$, every protected position fits and the binary role label separates protected content from filler. If $|S \cup M| > B$, the protected set is *oversubscribed*: a binary label cannot order positions within S, and the policy must resolve the excess by some other rule. We define the oversubscription ratio

$$\rho \;=\; \frac{|S \cup M|}{B}, \tag{3}$$

which is computable from the prompt before any token is generated.

The literature framing, and our own earlier one, stops here and treats $\rho > 1$ as the condition under which the prior ceases to be useful. That is incomplete. When $\rho > 1$ the policy does not fail arbitrarily; it falls back on whatever secondary ordering it implements. The evaluated policy admits protected positions in index order, so the surviving subset of S is its *earliest* elements. Whether the trace's load-bearing state is inside that surviving prefix is therefore a second, independent condition. Section 5 shows that this second condition, not ρ, decides which tasks break.

3.3 ADAPT: a budget-relative blend

A hard union of structural protection with attention-based selection consumes the whole budget whenever $\rho > 1$, and in earlier experiments performed no better than SnapKV alone and worse than both parents at tight budgets. ADAPT replaces the hard constraint with a budget-relative prior,

$$\alpha = \min\left(1, \frac{B}{|S|}\right), \qquad s = \alpha \cdot \mathrm{rank}(s_{\mathrm{struct}}) + (1-\alpha)\cdot \mathrm{rank}(s_{\mathrm{attn}}), \tag{4}$$

where both quantities are known from the prompt and no scalar is fitted to any outcome. Rank normalisation is required because the structural bands are of order 10^6 while attention scores are of order 10^{-3}; a raw weighted sum would be dominated by scale rather than by preference. At $\alpha = 1$ ADAPT recovers the structural ordering exactly; as oversubscription grows it shifts weight toward attention. ADAPT computes attention scores and is therefore strictly more expensive than SnapKV; it is evaluated for accuracy at aggressive compression, not as a cost reduction.

4 PriorityBench-A

PriorityBench-A generates multi-turn chats in which a designated span carries the state required by a final request, while intervening turns supply benign filler. Scoring is binary and programmatic, with no model judge. Three task families are represented equally (Table 1). Two frozen manifests serve different studies: a 240-example set spanning 8k, 16k and 32k nominal contexts for the storage study, and a 120-example stress set at 8k and 16k for retention.

Table 1: PriorityBench-A task families. All scorers are deterministic.

Family	Required behaviour	Scorer
Tool schema	Emit a valid call under an early contract	JSON/schema subset
Instruction supersession	Follow the latest conflicting constraint	Required/forbidden regex
Multi-turn state	Reuse an earlier identifier, path, or preference	Exact slot match

Because the generator's templates are visible to the tagger, a heuristic could appear semantic merely by recognising them. We therefore evaluate two transformations of each slice. *Middle relocation* moves state away from the prefix while holding the leading system message and the final request fixed. *Buried state* lengthens or embeds state-bearing turns so that a short-span heuristic cannot protect all of the gold. Separately, a CPU audit maps gold-bearing messages to tokenizer spans and intersects them with the selected positions, using no model outputs, to measure how much gold already lies within the always-kept window. On the stress slices only 1.0–1.3% of Qwen gold tokens and none of the Llama gold tokens fall inside sink plus recent, so the in-distribution gap over position-blind eviction reflects retention rather than a labelling leak.

4.1 In-distribution behaviour

At a 25% keep budget on the stress set, structural retention passes 112 of 120 Qwen examples against 1 of 120 for the position-blind control. We report this result but decline to build on it: the control retains only sink plus recent on a benchmark whose gold state is deliberately placed outside that window, and is therefore close to guaranteed to fail. The informative comparison is against attention-based selection, where structure passes 112/120 and SnapKV, PyramidKV, and a structure–SnapKV hybrid each pass 108/120; the paired structure–SnapKV difference is not significant (exact McNemar, $b = 4$, $c = 0$, $p = 0.125$). Under middle relocation, structure and FullKV both reach 115/120; under buried state, structure falls to 80/120 against FullKV at 107/120. Visible roles are thus a useful prior, not an oracle for arbitrarily embedded state.

5 Measuring the Operating Boundary

5.1 Design

To convert the boundary from an inference into a measurement we require traces whose protected-role mass varies continuously while everything else is held constant. We obtain them by converting filler user/assistant pairs into *distractor* tool-schema pairs: a short registry notice followed by a JSON dump of deferred schemas, sized to the characters it replaces. The gold task, its scoring payload, and the approximate context length are unchanged; only the composition of the surrounding context varies. Eight levels span 5.2%–92.2% measured protected-role mass, crossed with keep budgets $k \in \{0.02, 0.05, 0.10\}$ at 16k nominal context, with $n = 60$ examples per cell. The same base example appears at every level, so comparisons along the axis are paired on the task. Every non-FullKV arm is a `kvpress` press over an identical full prefill at an identical compression ratio, so the arms differ only in policy and never in mechanism.

Two earlier iterations of this design returned nulls, and both are informative enough to report. In the first, gold state occupied its natural position in the template prefix; every non-blind arm then scored 1.000 in all 24 cells, including at ninefold oversubscription. The cause is the tiebreak of Section 3.2: when the protected set does not fit, the policy retains its earliest members, which in that layout are the gold positions themselves. Oversubscription could not bite regardless of how large it became. In the second, gold was relocated to mid-context but the transform was applied after schema conversion; because the relocation helper classifies converted schema turns as non-filler, it bundled them with the gold block and inserted that block after a shrinking filler prefix. Gold

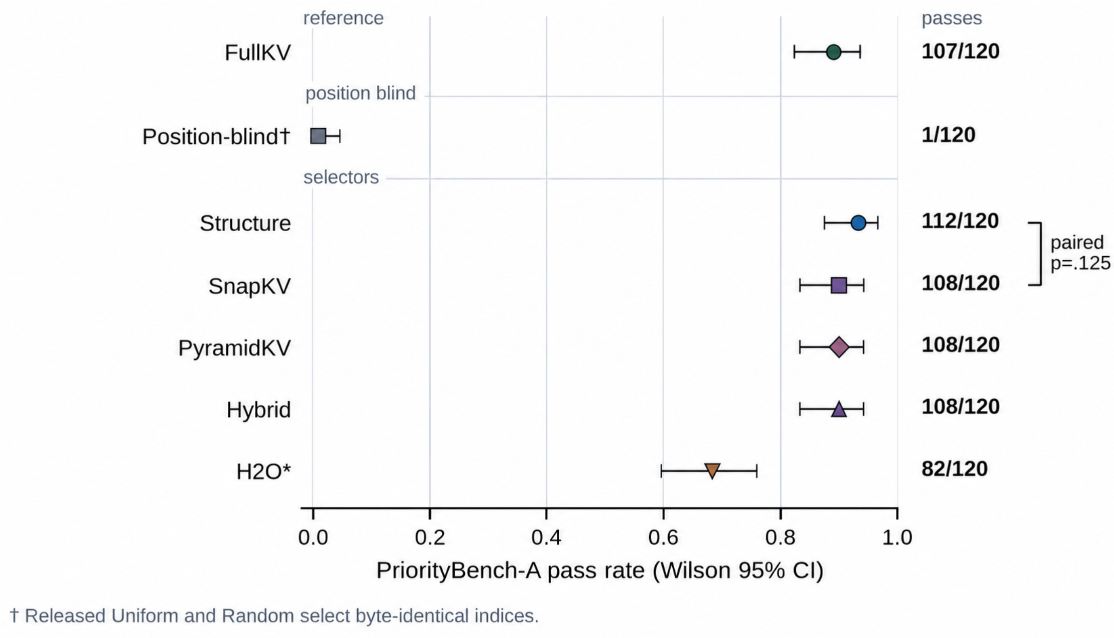

Figure 2: In-distribution comparison at a 25% budget ($n = 120$, Qwen3-8B). Points are pass rates with Wilson 95% intervals; the bracket reports the paired structure–SnapKV McNemar test. H2O* denotes a chunked reimplementation carrying a fidelity caveat. This operating point is the one at which structure and attention selection agree; Section 5.2 shows the agreement does not survive oversubscription.

consequently drifted back toward the start as protected mass rose—from character fraction 0.65 at the lowest level to 0.07 at the highest—cancelling the effect that the sweep was designed to expose. The reported sweep applies relocation *before* conversion, with gold verified at character fraction 0.63–0.65 at every level.

5.2 A sharp boundary that replicates

Structural retention is at ceiling while the protected set fits the budget and degrades immediately once it does not (Table 2, Figure 3). The knee lies near $\rho \approx 1.1$ on both model families. FullKV passes every cell, so the collapse reflects retention rather than task difficulty, and the position-blind arms remain near zero throughout.

Table 2: Pass rate against oversubscription ratio ρ, $n = 60$ per cell. Ranges span the cells falling in each band.

ρ	Structure (Qwen)	Structure (Llama)	SnapKV	ADAPT	FullKV
0.5×	1.000	1.000	1.000	1.000	1.000
1.0×	1.000	1.000	1.000	1.000	1.000
1.1×	1.000	0.883	1.000	1.000	1.000
2.2×	0.617	0.600	1.000	1.000	1.000
4–18×	0.400	0.333–0.417	0.967–1.000	1.000	1.000
27–45×	0.067	0.000	0.750–0.867	0.950–1.000	1.000

Past the boundary, the parity with attention-based selection reported in Section 4.1 reverses. On Llama, structure loses to matched-budget SnapKV in 20 of 24 cells, with $b = 0$ in every one—that is, structure wins no discordant pair anywhere. On Qwen the same pattern holds at the tightest budget, with $b = 0$ and c between 41 and 54. The in-distribution parity is therefore a property of one operating point rather than a general result, and should not be reported without it.

5.3 Which state survives, and why

Pooling across task families conceals the mechanism, and the concealment is not subtle: the pooled floor near 0.400 is the average of two families at 1.000 and 0.000 respectively. Table 3 separates them.

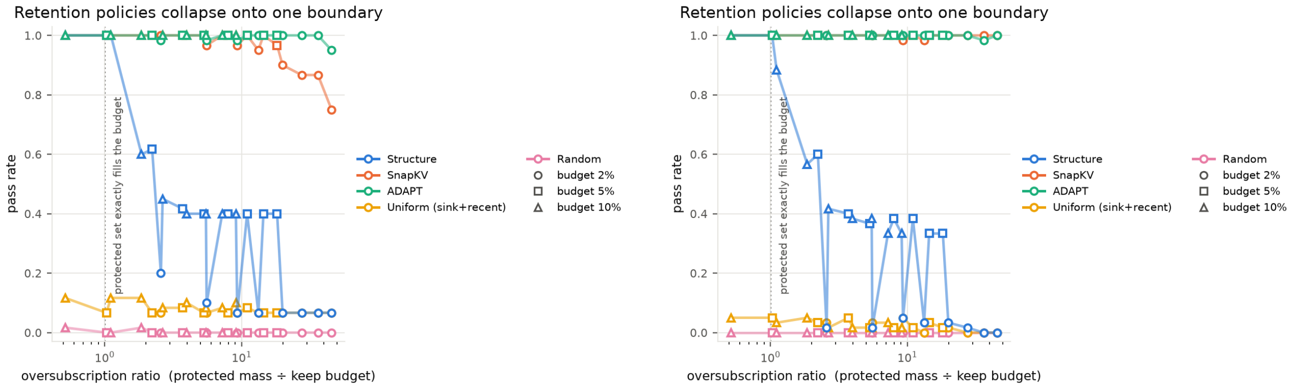

Figure 3: One boundary, two architectures. Pass rate against oversubscription ratio for Qwen3-8B (left) and Llama-3.1-8B (right). Marker shape encodes the keep budget, so points originating from 2%, 5% and 10% budgets can be seen to lie on a common curve. The dotted line marks the protected set exactly filling the budget.

Table 3: Structural retention by task family at high oversubscription ($\rho \geq 5$), $n = 20$ per cell per model.

Model	Budget	Tool schema	Supersession	Multi-turn state
Qwen	0.10	1.000	0.000	0.200
Qwen	0.05	1.000	0.000	0.200
Qwen	0.02	0.000	0.000	0.200
Llama	0.10	1.000	0.000	0.000–0.150
Llama	0.05	1.000	0.000	0.000–0.150
Llama	0.02	0.000	0.000	0.000–0.050

Tool-schema conformance survives eighteenfold oversubscription intact, while instruction supersession is extinguished. The explanation is structural rather than statistical. A tool-schema item's load-bearing content is the leading system block; the policy admits protected positions in index order; therefore the contract survives however oversubscribed the policy becomes, until the budget is too small to hold the block at all—which is precisely the $k = 0.02$ row, where tool-schema conformance also collapses. Supersession and multi-turn state live in mid-conversation turns, which the same ordering discards first.

This supersedes the account of Section 3.2 in an important way. The ratio ρ governs *whether* a binary structural label can still discriminate. The position of load-bearing state relative to the tiebreak governs *what is lost when it cannot*. Only the pair is predictive, and a serving policy consulting ρ alone would be systematically over-optimistic for exactly those workloads—agent traces with accumulating state—that motivate structural retention in the first place.

5.4 ADAPT across the range

ADAPT holds 0.950–1.000 on Qwen and 0.983–1.000 on Llama across all 24 cells per model, and loses no discordant pair to SnapKV in any of the 48 cells. Its measured α decays monotonically from 1.000 to 0.022 as oversubscription grows, with no value fitted to any outcome.

On Qwen at $k = 0.02$ it significantly exceeds SnapKV in four cells (Table 4), always with $c = 0$. The advantage does not replicate on Llama, where SnapKV never falls below 0.983 at any tested setting and therefore leaves no headroom to recover. We claim only that ADAPT matches attention-based selection everywhere tested and exceeds it where attention-based selection itself begins to degrade.

6 External Evaluation

6.1 The prior fails, as the mechanism predicts

BFCL V3 multi-turn provides an externally authored context distribution scored by an unmodified official checker that executes tool calls against stateful API implementations [11]. At a 25% keep budget over 140 paired Qwen

Table 4: ADAPT versus SnapKV on Qwen3-8B at $k = 0.02$, exact two-sided McNemar, $n = 60$ per cell.

Protected mass	ADAPT	SnapKV	b	c	p
0.399	1.000	0.900	6	0	3.1×10^{-2}
0.549	1.000	0.867	8	0	7.8×10^{-3}
0.723	1.000	0.867	8	0	7.8×10^{-3}
0.909	0.950	0.750	12	0	4.9×10^{-4}

conversations, FullKV reaches 0.193 and SnapKV 0.136, a difference that is not significant ($p = 0.152$); the hard structural policy scores 0.000, and the paired comparison against FullKV gives $p = 1.5 \times 10^{-8}$. The same ordering holds on 143 Llama conversations. BFCL prompts are dominated by tool schemas and carry 98.8% protected-role mass, so any tight budget is oversubscribed by more than an order of magnitude.

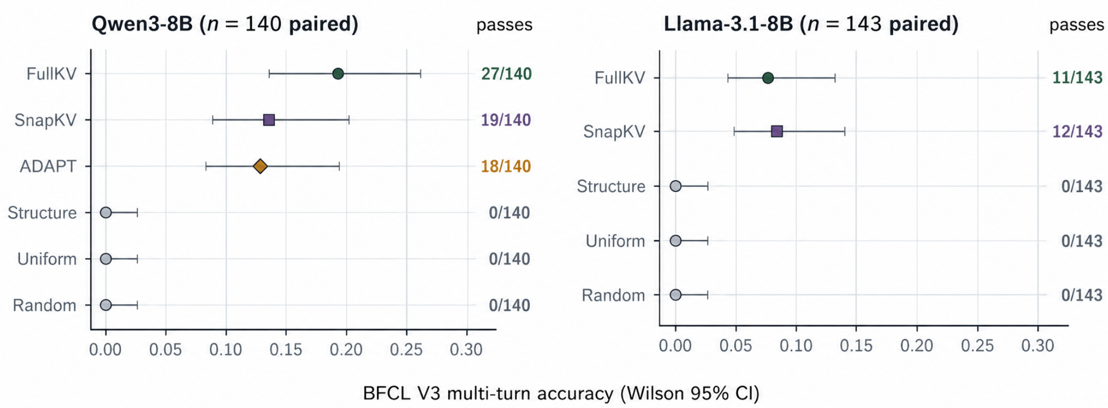

Figure 4: Paired BFCL accuracy with Wilson intervals. The ordering holds on both model families: no significant FullKV–SnapKV difference is detected, while the non-attention policies sit at zero with a Wilson upper bound of 0.027.

6.2 Why the workload cannot reward the prior

A census of failure modes on the frozen points explains the result more precisely than the ratio does, and does so without new generation (Table 5).

Table 5: BFCL failure modes by arm at $k = 0.25$, official checker error types.

Arm	Pass	Empty turn response	State mismatch	Execution mismatch
FullKV	20.5%	45.5%	22.7%	11.4%
SnapKV	13.9%	50.4%	21.2%	14.6%
ADAPT	12.9%	46.4%	27.1%	13.6%
Structure	0.0%	78.0%	10.0%	12.0%

Two facts follow. First, 45.5% of conversations fail with an empty or unparseable turn response *even with the full cache*, and FullKV passes only 20.5%; on this workload an 8B model is limited more by its own ability to emit well-formed multi-turn tool calls than by retention. The entire interval within which a retention policy can be judged is the roughly eight conversations separating SnapKV from FullKV. Second, the residual failures are dominated by state mismatch—loss of accumulated multi-turn state—which is precisely the class that Section 5.3 shows the structural prior does not protect. The schemas that it does protect sit in the system block and are evidently not the binding constraint. Consistent with this, the ADAPT arm shows a higher state-mismatch rate than SnapKV (27.1% versus 21.2%) at marginally lower accuracy.

6.3 A pre-registered test and its budget

The mechanism therefore yields a directional prediction, which we registered before scoring any conversation at a tight budget: on BFCL, ADAPT should match SnapKV within noise or fall slightly below it, because at this workload's protected mass the structural component biases retention toward schema tokens at the expense of the state the checker scores. The registration also fixes the analysis (exact McNemar paired on task, $n = 300$), the interpretation of every possible outcome, and a minimum detectable effect; with $c = 0$ the design detects $b \geq 6$, and a null bounds any true advantage below roughly 0.05 absolute.

Selecting the budget required care, because the regime in which ADAPT can help is one where attention-based selection is degraded but not extinguished. A pre-specified rule chose the tightest rung at which SnapKV falls between 0.25 and 0.70 of FullKV on the same conversations. A ladder of SnapKV-only pilots (Table 6) shows that attention-based selection floors on BFCL below a 10% budget, so the tight budgets at which the synthetic advantage appears do not exist on this workload; the rule selects $k = 0.15$.

Table 6: Budget-selection pilot, SnapKV only, against the budget-independent FullKV reference on the same conversations.

Budget	n	SnapKV	FullKV	SnapKV/FullKV
0.02	49	0.000	0.184	0.00
0.05	49	0.000	0.184	0.00
0.10	48	0.021	0.188	0.11
0.15	89	0.124	0.202	0.61
0.20	92	0.087	0.207	0.42

6.4 Outcome: the predicted null

The main test ran 300 conversations per arm at $k = 0.15$, yielding 233 fully paired across FullKV, SnapKV and ADAPT at 94% paired completeness, with zero matched-budget violations and 33 exclusions, all of them context-limit exclusions distributed across arms (Table 7).

Table 7: BFCL V3 multi-turn at the pre-registered budget $k = 0.15$, official checker, $n = 233$ fully paired conversations. Intervals are Wilson 95%; tests are exact two-sided McNemar.

Arm	Passes	Accuracy	Wilson 95%
FullKV	52/233	0.223	[0.174, 0.281]
SnapKV	32/233	0.137	[0.099, 0.187]
ADAPT	26/233	0.112	[0.077, 0.158]
Paired comparisons			
ADAPT vs SnapKV	b=10, c=16	$\Delta = -0.026$	$p = 0.327$
SnapKV vs FullKV	b=12, c=32	$\Delta = -0.086$	$p = 0.0037$
ADAPT vs FullKV	b=12, c=38	$\Delta = -0.112$	$p = 0.0003$

ADAPT is statistically indistinguishable from SnapKV and trends 2.6 points below it. This is the registered prediction, and the observed effect lies well inside the bound the registration specified: a null on this design constrains any true ADAPT advantage to below roughly 0.05 absolute, and the measured Δ is -0.026 with a 10/16 discordant split. The mechanism of Section 5.3 therefore made a quantitative, directional, pre-registered prediction about a benchmark the authors did not construct, and the prediction was borne out.

We emphasise what this does and does not establish. It does not establish that the method is useful on BFCL; it establishes the opposite, and the paper claims no external benefit. What it supports is the *mechanism*: a structural prior that protects system-block content cannot help a workload whose residual failures are losses of accumulated multi-turn state, and the size and direction of that shortfall were derivable in advance from a synthetic sweep. The two secondary comparisons are consistent with the same account. SnapKV trails FullKV significantly ($p = 0.0037$), confirming that a 15% budget is genuinely lossy on this workload rather than free, and ADAPT trails FullKV more strongly ($p = 0.0003$), as expected when 98.8% protected mass oversubscribes the budget more than sixfold.

6.5 A generation-free corroboration

An audit of 4,856 public τ-bench trajectories (828,000 extracted spans), using no model outputs, corroborates the mechanism from the retention side: structural retention preserves explicit policy lines at 0.820 against 0.001 for a position-only budget, but loses on reused identifiers (0.055 against 0.315). The prior wins precisely where protected spans are scarce and separable, and loses on the recently referenced values that a recency heuristic captures.

7 Storage: A Matched-Precision Negative

We tested separately whether role-awareness helps when positions are retained at reduced precision rather than evicted. Holding the INT4 fraction at exactly 0.75 and the sink/recent protection common, role-aware placement and uniform placement are indistinguishable: over 240 locked examples FullKV passes 213, uniform-mixed 211, and structure-mixed 212, a one-example difference that no paired test supports. We report this as a negative result.

The implemented mixed BF16/INT4 path does reduce stored bytes, at a measured cost. Payload falls to 0.719× FullKV and allocated peak to 0.868×, while time per output token rises by 1.20–1.215× and end-to-end latency by 1.11–1.12×. The gap between the 0.719× measured payload and an idealised 0.473× byte model is accounted for by storing each 4-bit code in a `uint8` container with FP32 scale and zero-point metadata; the gap between payload and peak is accounted for by dequantising cold pages into BF16 scratch for attention. These are engineering costs of the prototype rather than properties of the idea, but they are the numbers this implementation achieves, and the latency comparison additionally spans two different attention backends.

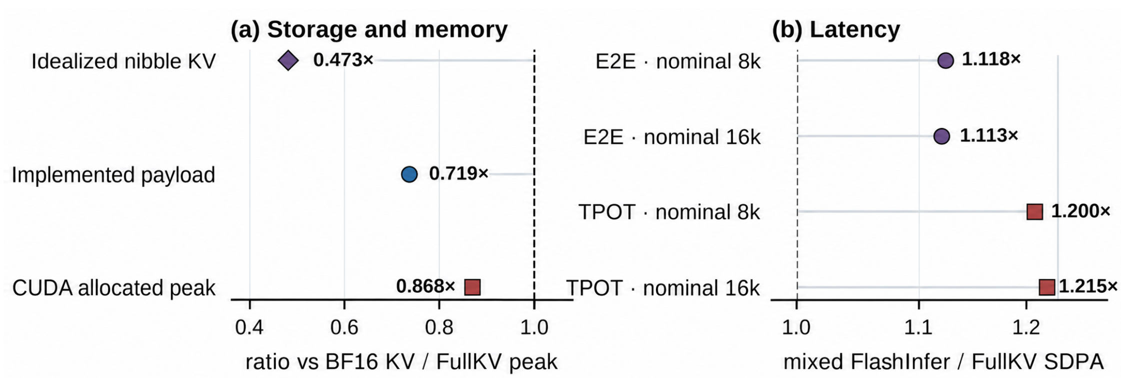

Figure 5: Storage trade-off on H200. The implemented code-plus-metadata payload is smaller than the FullKV cache, while the allocated peak falls less because cold pages are expanded into BF16 scratch for attention. Latency compares the mixed FlashInfer path against FullKV SDPA; the dashed line marks parity.

8 Discussion

A cheap signal with a knowable range. The practical value of a structural prior lies in what it does not require: no attention statistics, no model behaviour, nothing beyond the prompt. Our results delimit where that suffices. While the protected set fits the keep budget, structural retention matches attention-based selection on the tasks we can measure; past that point it degrades sharply and loses to it decisively. Because ρ is computable before generation, the choice between the free signal and the paid one can be made per request.

But the ratio is not the whole diagnostic. The dissociation of Section 5.3 shows that ρ predicts whether the label can discriminate, not what is lost when it cannot. A workload whose critical state sits in the system preamble tolerates extreme oversubscription; a workload whose critical state accumulates across turns does not tolerate even modest oversubscription. Agent traces are largely of the second kind, which is an uncomfortable finding for the premise that motivated this line of work, and it is the reason the same prior that succeeds on schema-conformance tasks fails on BFCL.

On the value of soft weighting. ADAPT never loses to attention-based selection across the tested grid and significantly exceeds it where that selection begins to degrade. Because it consumes attention scores, it cannot be justified as a cost reduction; its case rests on accuracy at aggressive compression, and that case is currently established on one model family in one regime.

9 Limitations and Threats to Validity

The positive results are on a benchmark we authored. Every result showing an advantage is measured on PriorityBench-A. The externally authored evaluation shows the hard structural policy at 0.000, and the pre-registered follow-up measured no ADAPT benefit there ($\Delta = -0.026$, $p = 0.33$). No claim of external superiority is made or supported, and the reader should treat the ADAPT advantage of Section 5.4 as an in-distribution result whose external counterpart has been sought and not found.

Instrument and stimulus are not independent. The tagger keys on role, length, and schema or constraint markers that resemble the generator's visible templates, and the sweep's distractor schemas were authored to carry roles the same tagger recognises. The protected-mass axis is measured post hoc with the pinned tokenizer and the mechanism replicates across two model families, but this does not eliminate the concern, and it is the principal reason the external evaluation rather than the sweep must carry any method claim.

No difficulty gradient in the sweep. FullKV passes every cell of the sweep. This isolates retention cleanly but means the benchmark cannot speak to workload realism, and that pass rates are close to binary.

Scope of the ADAPT result. The advantage over SnapKV is confined to Qwen3-8B at a 2% budget. On Llama the comparison is uninformative rather than negative, because SnapKV does not degrade enough to leave headroom.

External benchmark headroom. On BFCL, an 8B model passes only 20.5% of conversations with the full cache and fails 45.5% with unparseable turn responses. The workload therefore adjudicates retention policies weakly, and our external conclusions are correspondingly bounded.

Model, context, and budget coverage. Two 8B models, nominal contexts to 32k, and selected budgets between 2% and 50% are tested. Behaviour at larger scales, longer contexts, and under batched serving is unmeasured. Systems measurements are single-request and do not establish throughput, concurrency, or tail latency.

Relation to concurrent work. Role-aware retention is an active area, and we do not claim the use of structural roles as a retention signal. RoleKV [12] predates this work on the signal itself, and [13] predates it on the existence of a density-dependent boundary, albeit at the sparse end of that axis and over document-structural rather than conversational roles. Our claims are confined to the saturated regime, to lossy eviction, and to the tiebreak interaction of Section 5.3, which we believe to be new but which a reader should weigh against that literature rather than in isolation.

Baseline fidelity. The sweep and external evaluations use the `kvpress` implementation of SnapKV. The earlier in-distribution comparisons additionally use repository reimplementations of PyramidKV and a chunked H2O; the latter carries an explicit fidelity caveat and should not be read as a reference-H2O result.

10 Conclusion

Application-visible structure is a retention signal that costs nothing to compute, and this paper establishes the range over which it is sufficient. Across 15,360 scored generations on two model families, structural retention holds at ceiling while the protected set fits the keep budget and degrades sharply beyond an oversubscription ratio of approximately one; past that point it loses to matched-budget attention-based selection without winning a single discordant pair. The boundary is computable before generation, which makes it usable as a serving-time diagnostic.

The more consequential finding is that the ratio alone is not that diagnostic. Because the policy resolves ties by position, state carried in the leading system block survives arbitrary oversubscription while state accumulated across mid-conversation turns is lost entirely, and it is this second factor that determines whether a workload is served by the cheap signal. The distinction is not a refinement of the earlier account but a correction of it, and it converts an unexplained external failure into a predicted one. That prediction was registered before the corresponding experiment was scored and the experiment returned it: on 233 paired conversations of an externally authored agent benchmark, the blended policy is indistinguishable from attention-based selection and trends slightly below it, as the mechanism requires. The method does not transfer. The account of why it does not transfer does.

In operational terms: keep the schema before the pleasantry, but check first whether the budget can hold the schema—and whether the state your task actually depends on is the schema at all.

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